Orion S. Ning

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Education

The University of California, Berkeley — Berkeley, CA
Ph.D. Physics, NSF Graduate Research Fellow — *2022 –*

The University of California, Berkeley — Berkeley, CA
B.A. Physics (Highest Honors), Mathematics (High Honors) — *2022*
M.A. Physics — *2024*

Research Interests

Dark Matter Phenomenology and Detection, Axions, Astrophysics/Astroparticle Physics, Computational Methods, Quantum Field Theory

Research

Dark Matter Indirect Detection and Phenomenology — 2022 –
Advisor: Benjamin R. Safdi — *UC Berkeley, and LBNL*
Funding: NSF Graduate Research Fellow

Publications

INSPIRE |

1. **Ning, O.**, Safdi, B.R., & Welch, C. (2026). No Evidence for Superradiant Axions in LIGO-Virgo-KAGRA GWTC-5 Binary Black Hole Spins. arXiv:2607.01317.
2. Elllis, S. A. R., **Ning, O.**, Rodd, N. L., & Schütte-Engel, J. (2026). Crowdsourcing Gravitational Waves from Superradiant Axions. arXiv:2603.15734.
3. De Marchi, A. G., **Ning, O.**, & Xiao, T. (2026). Blazar Constraints on Axions through New Spectral Modulation Searches in 1ES 1959+650 & B2 1811+31. arXiv:2603.13480.
4. **Ning, O.**, Raman, K., & Safdi, B. R. (2025). Cosmological Neutron Stars Produce Diffuse Axion X-Ray Signatures. arXiv:2512.15849.
5. **Ning, O.**, Ray, A., & Safdi, B. R. (2025). Axion lines from nuclear de-excitations in galactic stellar populations. *Phys.Rev.D 113 (2026) 3, 035010* arXiv:2509.03569.
6. **Ning, O.**, & Safdi, B. R. (2025). Probing the Axion-Electron Coupling with NuSTAR Observations of Galaxies. *Phys.Rev.D 113 (2026) 1, 015021* arXiv:2503.09682.
7. **Ning, O.**, Dessert, C., Hong, V., & Safdi, B. R. (2024). Search for Axions from Magnetic White Dwarfs with Chandra. *Phys.Rev.D 111 (2025) 10, 103002* arXiv:2411.05041.
8. **Ning, O.**, & Safdi, B. R. (2024). Leading Axion-Photon Sensitivity with NuSTAR Observations of M82 and M87. *Phys.Rev.Lett. 134 (2025) 17, 171003* **Editor's Suggestion,** arXiv:2404.14476.
9. Dessert, C., **Ning, O.**, Rodd, N. L., & Safdi, B. R. (2023). Resurrecting Hitomi for Decaying Dark Matter and Forecasting Leading Sensitivity for XRISM. *Phys.Rev.Lett. 132 (2024) 21, 211002* arXiv:2305.17160.
10. Mu2e-II Collaboration (including **Ning, O.**) (2022). Mu2e-II: Muon to electron conversion with PIP-II. *In Snowmass 2021.* arXiv:2203.07569.
11. Garcia-Quintero, C., Ishak, M., & **Ning, O.** (2020). Current constraints on deviations from General Relativity using binning in redshift and scale. *JCAP, 12, 018.* arXiv:2010.12519.

Selected Presentations, Talks, and Poster Sessions

- "Dark Particles and Bright Bursts: Future Stories from the Gamma-Ray Sky with GALAXIS," Closing In on Thermal Relics with Indirect, Direct, and Collider Searches for Dark Matter Conference (CERN), *Geneva, Switzerland, July 2026*
- "Axion Searches from Black Hole Superradiance," CERN Theory Cosmo/BSM Coffee, *Geneva, Switzerland, July 2026*
- "Crowdsourcing Gravitational Waves from Superradiance: Black Holes as Extreme Axion Laboratories," 4th General Meeting of COST Action COSMIC WISPers CA21106, *Corfu, Greece, June 2026*
- "Dark Particles and Bright Bursts: Future Stories from the Gamma-Ray Sky with GALAXIS," COST WG3 (WISPs in Astrophysics) Monthly Meeting, *online, April 2026*
- "Dark Particles and Bright Bursts: Future Stories from the Gamma-Ray Sky with GALAXIS," LBL Dark Matter Lunch Talk, *Berkeley, CA, April 2026*
- "Galactic Axion Factories: Astrophysical Populations and Signals for Axion Searches," Stanford KIPAC Seminar, *Stanford, CA, March 2026*
- "Galactic Axion Factories: Stellar Populations and X-Ray Signals for Axion Searches," SLAC Theory Seminar, *Stanford, CA, October 2025*
- "Dark Particles and Bright Bursts: Future Stories from the Gamma-Ray Sky with GALAXIS," UC Berkeley Astrophysics Department Lunch Talk, *Berkeley, CA, October 2025*
- "Galactic Axion Factories: Stellar Populations and X-Ray Signals for Axion Searches," Axions in Stockholm 2025, *Stockholm, Sweden, July 2025*
- "Galactic Axion Factories: Stellar Populations and X-Ray Signals for Axion Searches," The Strong CP Problem and its Possible Solutions 2025, *Pollica, Italy, June 2025*
- "Galactic Axion Factories: Stellar Populations and X-Ray Signals for Axion Searches," Berkeley 4D Seminar, *Berkeley, CA, April 2025*
- "Axion-Photon Sensitivity with NuSTAR Observations of M82 and M87," TeVPA 2024, *Chicago, IL, August 2024*
- "Axion-Photon Sensitivity with NuSTAR Observations of M82 and M87," DPF-PHENO 2024, *Pittsburgh, PA, May 2024*
- "Resurrecting Hitomi for X-ray Probes of Sterile Neutrino and Axion-Like-Particle Dark Matter with Blank Sky Observations," PASCOS 2023, *Irvine, CA, June 2023*
- "Compactification of a Nonlocal 6D Quantum Field Theory Toward Exploring Anomalies," UC Berkeley Summer Undergraduate Research Fellowship (SURF) Conference, *Berkeley, CA, August 2021*

Teaching and Tutoring

Teaching Assistantships (GSI)

- Physics 110B: Classical Electrodynamics Part II *(instructor: Liang Dai)* ..... Fall 2023

Democratic Education at Cal (DeCal) | *Lead Course Facilitator and Teacher* site

- Astronomy 98: Beginner's Guide to the Universe *(faculty sponsor: Jessica Lu)* ..... Fall 2020 – Spring 2022

Department Peer Tutor and Group Study Facilitator | *Physics Department* site

- Physics 137A (Quantum Mechanics I) ..... Spring 2022

- Physics 110A (Electromagnetism and Optics I) ..... Fall 2021
- Physics 137B (Quantum Mechanics II) ..... Spring 2021, Fall 2020

Course Reader/Grader

- Physics 209: Classical Electromagnetism (Graduate Level) *(instructor: Liang Dai)* ..... Fall 2024
- Physics 137A: Quantum Mechanics I *(instructor: Swapan Chaterjee)* ..... Summer 2023
- Physics 110B: Classical Electrodynamics Part II *(instructor: Ehud Altman)* ..... Spring 2022
- Physics 112: Statistical Mechanics/Thermodynamics *(instructors: Michael Zaletel, Ehud Altman)* ..... Fall 2021
- Physics 137A: Quantum Mechanics I *(instructor: Dan Kasen)* ..... Spring 2021
- Physics 77: Introduction to Computational Techniques in Physics *(instructor: Yury Kolomensky)* ..... Fall 2020
- Math 54: Linear Algebra and Differential Equations *(instructor: Rui Wang)* ..... Summer 2020

Pedagogy Theory | *Courses Taken*

- MPS 375: Pedagogy in Math and the Physical Sciences ..... Spring 2023
- Student Learning Center (SLC) Astronomy and Physics Pedagogy Seminar ..... Fall 2021 – Spring 2022

Outreach, Service, and Mentoring

Berkeley Center for Theoretical Physics 4D Seminar site

- Organizer ..... Spring 2026

Astronomy on Tap San Francisco site

- Speaker, "From Astrophysics to Axions: The Glorious Hunt for Dark Matter" (Dec. 9) ..... Fall 2025

UC Berkeley Compass Lectures site

- Speaker, "The Cosmic Game of Hide-and-Seek: Chasing Dark Matter Across the Stars" (Apr. 18) ..... Spring 2025

UC Berkeley Astro Night Public Lecture site

- Speaker, "From Astrophysics to Axions: The Glorious Hunt for Dark Matter" (Oct. 3) ..... Fall 2024

Splash at Berkeley | *Administrator and Volunteer Teacher* site

- Administrator and Organizer ..... Spring 2024 –
- Class S1663: Cosmic Ghostbusters: Hunting the Universe's Invisible Beasts *(joint with Hannah Lykes)* Spring 2026
- Class S1570: Good Vibrations: Gravitational Waves and the Hidden Music of Spacetime ..... Fall 2025
- Class S1469: The Usual (Particle) Suspects: Culprits Beyond the Standard Model? ..... Spring 2025
- Class S1241: From Astrophysics to Axions: The Glorious Hunt for Dark Matter ..... Spring 2024, Fall 2024
- Class S919: So Long, and Thanks for All the Spacetime *(joint with Druv Punjabi)* ..... Spring 2022
- Class S838: Alge-cada-bra: The Algebraist's Answer to Life, the Universe, and Everything *(joint with Ryan Tamura)* ..... Fall 2021
- Rainstorm Virtual Class S732: The Quantum Complex *(joint with Ryan Tamura)* ..... Spring 2021
- Class S772: The Quantum Complex *(joint with Ryan Tamura)* ..... Spring 2021
- Class S667: Relativity and Spacetime ..... Fall 2020

Science Olympiad

- Science Olympiad at Berkeley: Exam Writer, Moderator, Grader – Solar System ..... Spring 2026, site
- Golden Gate Science Olympiad: Volunteer, Moderator & Grader – Astronomy ..... Spring 2025, site

UC Berkeley Major Curriculum Committee

- Graduate Representative ..... Fall 2022 – Spring 2023

- Undergraduate Representative ..... Fall 2021 – Spring 2022

MPS Scholars Mentoring Network | *Volunteer Mentor* site

- Mentor ..... Fall 2025 –

Berkeley Undergraduate Mathementoring Program (BUMP) | *Volunteer Mentor* site

- Mentor ..... Fall 2019 – Fall 2021

Society of Physics Students (SPS) Mentoring Program | *Volunteer Mentor* site

- Mentor ..... Fall 2020 – Spring 2022

Awards, Honors, and Funding

- Sushant Singh Rajput Memorial Fellowship in Physics 2025
- Hearts to Humanity Eternal (H2H8) Graduate Research Grant (site) 2025
- Outstanding Graduate Student Instructor (OGSI) Award (site) ..... 2023 – 2024
- National Science Foundation Graduate Research Fellowship (site) ..... 2022 – 2025
- Percy Lionel Davis Award in Mathematics (site) 2022
- Cal Alumni Association's The Leadership Award and Scholarship (site) ..... 2021 – 2022, 2018 – 2019
- Astronaut Scholarship Foundation (ASF) Scholar (site) ..... 2021 – 2022
- UC Berkeley Summer Undergraduate Research Fellowship (SURF) (site1) (site2) ..... Summer 2021
- Berkeley Physics Undergraduate Research Scholars Program (BPURS) Funding (site) ... Fall 2020, Spring 2021
- UC Berkeley Kraft Award for Freshmen (site) 2018

Technical Skills

Languages and Software: Python, C/C++, ROOT (CERN), Linux/Unix, BASH, cluster computing (slurm), Mathematica, LaTeX, Git, MESA, NSCool, CAMB/CosmoMC, Geant4 toolkits, MEGAlib, Arduino, LabVIEW, Microsoft Office Suite
Other: Electronics and Circuitry, Soldering